

CV-101 - Garage conveyor (Conv_Simple)

MIRA Print Pack - Verified Machine Print Pack

Approval tier: APPROVABLE WITH FIELD VERIFICATION

This pack documents CV-101 only. 14 devices · 40 modeled conductors (37 FIELD VERIFY) · 96 modeled terminals (34 FIELD VERIFY) · 28 open items (2 closed).

UNS path: enterprise.garage.demo_cell.cv_101

Customer: FactoryLM (bench/demo asset)

Site: Garage demo cell

As of: 2026-07-11

DISCLAIMER

This pack is provided as-is, evidence-graded at the tier stated on the cover page. Every FIELD VERIFY item is an explicit, located gap, not a confirmed fact - verify it on the physical machine before relying on it for energized work. This document is not a certified as-built, is not an arc-flash/short-circuit/NEC compliance study, and carries no control-write capability. The preparer is not liable for reliance on any item still marked FIELD VERIFY.

Questions - contact prints@factorylm.com.

E-001 COVER / LEGEND / DEVICE SCHEDULE

Conv_Simple / CV-101 (garage conveyor) — device schedule · sheet index · wire-numbering key · line-style legend

DEVICE SCHEDULE

Tag	Type	Model	Role	Evidence
PLC1	plc	Allen-Bradley Micro820 2080-LC20-20QBB	Conveyor controller; Modbus RTU master to VFD1	verified
VFD1	vid	AutomationDirect GS10 DURApulse	Drives conveyor motor. Power input is 230 V single-phase (R/L1, S/L2; T/L3 un...	verified
M1	motor		Conveyor drive motor — 230 V 3~ from VFD1 U/V/W (voltage technician-confirmed...	field_verify
S0	e stop		Emergency stop, DUAL-CHANNEL (diverse) — NC ch to I-02, NO ch to I-03	verified
SS1	selector switch		Direction selector — FWD contact to I-00, REV contact to I-01 (FWD-OFF-REV)	verified
S2	pushbutton (NO)		Run / rearm pushbutton to I-04; illuminated — lamp fed by PLC O-03 (PBRunLED)	verified
B1	photo eye		Product-present beam to I-05 (blocked -> pe_latched soft stop)	verified
PS1	power supply	Mean Well DIN-rail supply (24 V / 1.0 A; 100-24...	24 VDC control supply (E-004). Feeds the DC +/- distribution block (blue = on...	verified
DB1	distribution block		24 VDC +/- distribution block (push-in, WAGO-style, 2-level), fed from PS1's ...	verified
CB1	circuit breaker		2-pole branch breaker on the 230 V single-phase supply (existence/type/rating...	field_verify
Q1	control relay	Schneider Electric TeSys CA3KN22BD (control rel...	Control relay (Schneider CA3KN22BD, hand-labeled 'MLC') USED AS THE VFD SUPPL...	verified
PL1	pilot light		Green RUNNING pilot — load of O-00 (LightGreen)	verified
PL2	pilot light		Red FAULT/E-STOP pilot — load of O-01 (LightRed)	verified
X1	terminal block		Field terminal strip — field devices land here before PLC (E-008)	field_verify

SHEET INDEX

Id	Title	Status
E-001	Cover / legend / device schedule	drafted
E-002	Power one-line	drafted
E-003	VFD power (CB1 -> VFD1 -> M1)	drafted
E-004	24 VDC control power distribution (PS1)	drafted
E-005	PLC digital inputs	drafted
E-006	PLC outputs (O-00..O-06)	drafted
E-007	RS-485 Modbus (PLC1 Ch2 <-> VFD1 RJ45)	drafted
E-008	Terminal strip (X1) + wire list	drafted
E-009	Open items / field verification	drafted

WIRE-NUMBERING KEY

Conductor numbers are W[sheet-digit][line-2d] (e.g. W503 = sheet E-005, line 3) — all PROPOSED until as-built (OI-03/OI-19). Rail/bus labels (W24, W0V) are exempt. Comms sheets (E-007) use mnemonic conductor labels (485+/485-/SGND/SH) per CommsToVFD ancestry. Same electrical node = same wire number; a number changes only through a device.

LINE-STYLE LEGEND

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxxx

proposed wire number (red = unverified)

SAFETY

- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1 (style law §7; docs/references/industrial-wiring-diagram-standards.md).

NOTES

- LINE-STYLE LEGEND, red: red marks EITHER FIELD-VERIFY status (a dashed, unverified conductor/device or an unverified wire-number tag) OR a safety-critical prohibition (a NOT/never instruction inside a sheet's SAFETY block, e.g. 'a monitored input is NOT a safety stop') — same color, two distinct meanings distinguished by context (dashed line vs. SAFETY-block bullet). This is intentional package-wide convention, not per-sheet drift.

COVER / LEGEND / DEVICE SCHEDULE

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

cover generated from model/*.yaml

SHEET

E-001

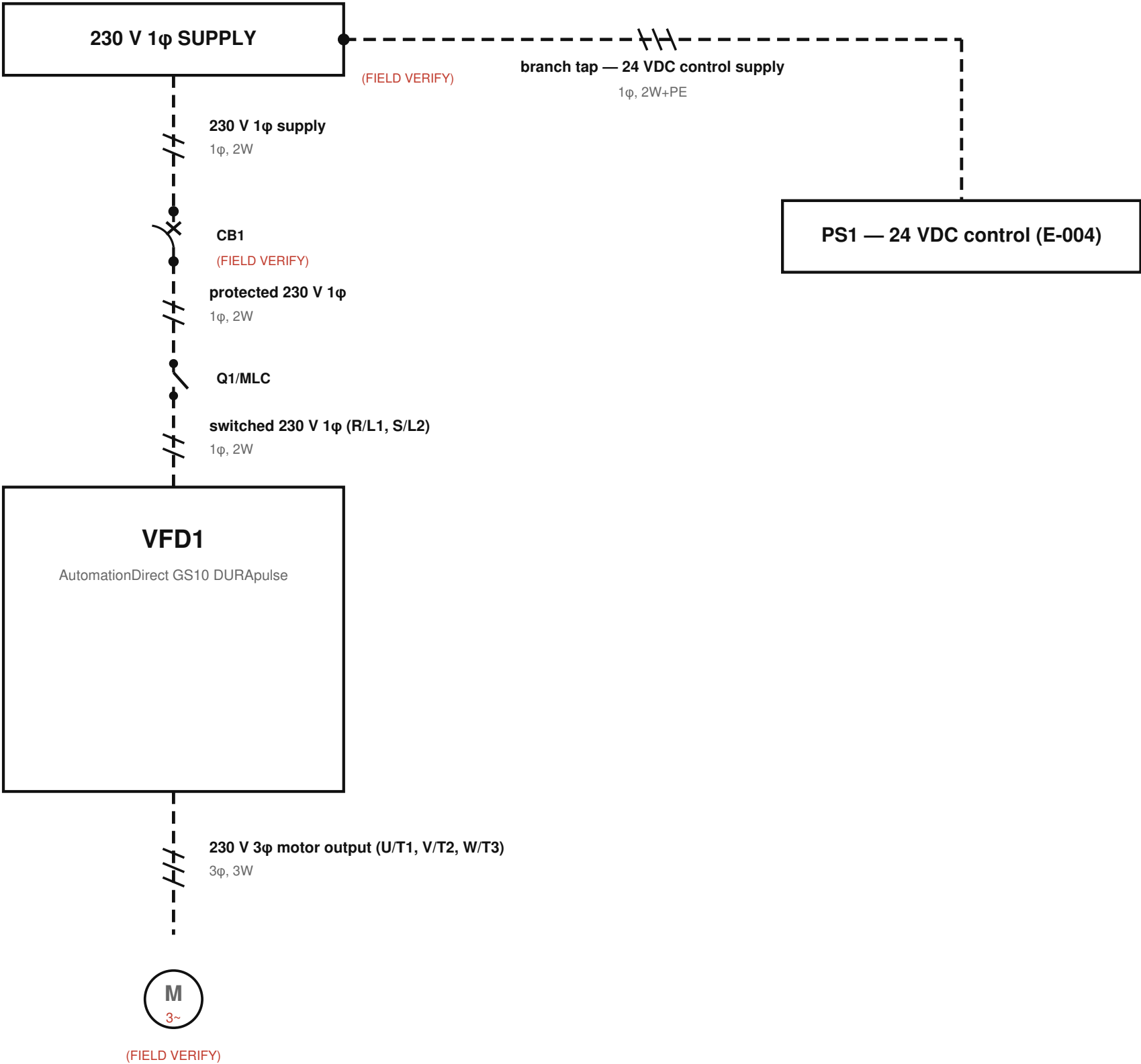
REV A

2026-07-11

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E-002 POWER ONE-LINE

Conv_Simple / CV-101 (garage conveyor) — 230 V 1φ supply → CB1 → Q1/MLC → VFD1 (GS10) → M1 · branch tap → PS1 (24 VDC control, E-004)



ONE-LINE CONVENTION

Each tick mark across a run = one conductor (e.g. 2 ticks = 2-wire single-phase; 3 ticks = 3-wire three-phase). Solid = verified, dashed = FIELD VERIFY, same as every other sheet (see the line-style legend). No per-conductor wire numbers on this sheet — see wires.yaml `convention` (E-001) for the numbered detail carried on E-003 / E-004.

Summary sheet only — no wire numbers are assigned here (see the wire-numbering key on E-001). For per-conductor evidence and FIELD VERIFY status, use E-003 (VFD power) and E-004 (24 VDC control power).

NOTES

- Conductor-count ticks (e.g. 2 ticks = '1φ, 2W') mark circuit conductor count only — not a conduit/cable schedule.
- Supply is 230 V SINGLE-PHASE (2-wire), technician-confirmed 2026-07-11 — this sheet corrects the earlier 3-phase assumption.
- PS1 branch is drawn as a tap off the same incoming supply; the exact AC leg/circuit feeding PS1 is FIELD VERIFY (OI-25).

SOURCES:

model/e002_online.yaml · review/PHOTO_EVIDENCE_V6.md (technician field correction 2026-07-11)

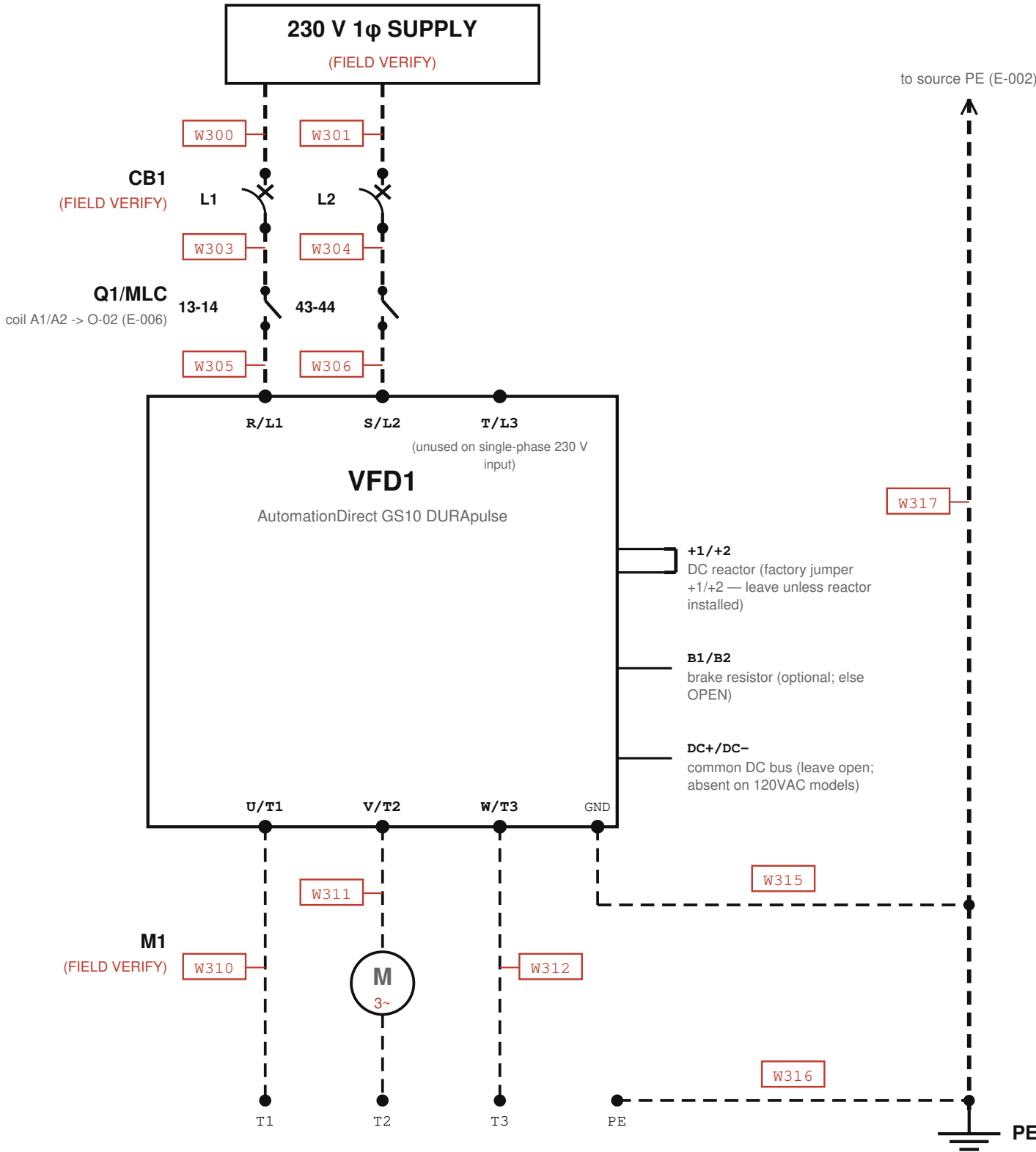
LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxxx proposed wire number (red = unverified)

POWER ONE-LINE Conv_Simple / CV-101 (garage conveyor)	SHEET E-002
Drawn: MIRA / FactoryLM one-line generated from model/e002_online.yaml	REV A 2026-07-11 2 of 9

E-003 VFD POWER

Conv_Simple / CV-101 (garage conveyor) — supply → CB1 → Q1/MLC (supply switch) → VFD1 (GS10) → M1 · 230 V 1φ in → 230 V 3φ to motor · every conductor FIELD VERIFY



Supply is CONFIRMED 230 V single-phase (technician-confirmed 2026-07-11, review/PHOTO_EVIDENCE_V7.md) — R/L1, S/L2 only; T/L3 unused. Only the GS10 exact model/frame, breaker rating, and wire gauge remain NOT DOCUMENTED — every conductor FIELD VERIFY. P00.01 = 1.60 A (2026-05-20 export) is the only sizing clue. MLC (Q1) is a control relay USED AS the single-phase drive-supply switch — its 2 NO contacts (13-14, 43-44) carry the 230 V 1φ supply into the GS10; coil A1/A2 from O-02 (E-006). Energizing O-02 powers the drive (technician-confirmed 2026-07-11).

- SAFETY**
- Never start/stop via input power (L1811-1813); CB1 is branch protection only, not a run/stop device.
 - LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
 - Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1.
 - PE resistance ≤0.1Ω (L1760-61); shield/conduit bonded both ends (L1792-95); RFI jumper in/out FIELD VERIFY (L1693-1718, OI-17).

- LEGEND**
- VERIFIED (solid)
 - - - FIELD VERIFY (dashed)
 - Wxxx proposed wire number (red = unverified)

CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W300	SUPPLY (E-002)	CB1.1	L1 230V 1ph supply	power_line	field_verify	
W301	SUPPLY (E-002)	CB1.3	L2 230V 1ph supply	power_line	field_verify	
W303	CB1.2	Q1.13	L1 protected -> MLC	power_line	field_verify	
W304	CB1.4	Q1.43	L2 protected -> MLC	power_line	field_verify	
W305	Q1.14	VFD1.R/L1	L1 switched -> drive	power_line	field_verify	
W306	Q1.44	VFD1.S/L2	L2 switched -> drive	power_line	field_verify	
W310	VFD1.U/T1	M1.T1	motor phase U	motor_lead	field_verify	
W311	VFD1.V/T2	M1.T2	motor phase V	motor_lead	field_verify	
W312	VFD1.W/T3	M1.T3	motor phase W	motor_lead	field_verify	
W315	VFD1.GND	PE bus	drive PE (≤0.1Ω)	ground	field_verify	
W316	M1.PE	PE bus	motor frame PE	ground	field_verify	
W317	PE bus	SUPPLY (E-002)	PE to source	ground	field_verify	

- NOTES**
- CB1 REQUIRED per GS10_UM L1758-1759 — type/rating unknown (OI-15).
 - M1: swap any two motor leads to reverse — GS10_UM L1773-1776.
 - VFD control source = Modbus — see E-007. No control wiring on this sheet.
 - Supply is 230 V SINGLE-PHASE (2-wire); drive output = 230 V 3φ to a 230 V motor (technician-confirmed 2026-07-11).
 - GS10 manual also warns against cycling a POWER-CIRCUIT switching device for normal run/stop — emergency situations only (GS10_UM.txt L1754-1757), same caution as L1811-1813 above. This applies to Q1/MLC (the device that powers the drive) exactly as it applies to CB1: normal run/stop is via Modbus (E-007), never by cycling Q1's coil.

- SOURCES:**
- plc/GS10_UM.txt (terminals L1971-1986; topology L1750-1813) ·
 - plc/GS10_Integration_Guide.md (Q1 phase-5 test)
 - plc/Prog_init_ConvSimple_v2.1.st (vfd_run_permit) · plc/MIRA_PLC_WorkInstruction_v3.pdf (LOTO §2)
 - photos review/photos/wire_1.4.jpg (2026-07-11, actual devices)
 - review/PHOTO_EVIDENCE_V6.md (technician field correction 2026-07-11 — single-phase supply topology, MLC in power path)
 - review/PHOTO_EVIDENCE_V7.md (technician confirmation 2026-07-11 — 230 V motor output, OI-27 resolved)

VFD POWER

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM
terminals per GS10 UM 1st Ed Rev B

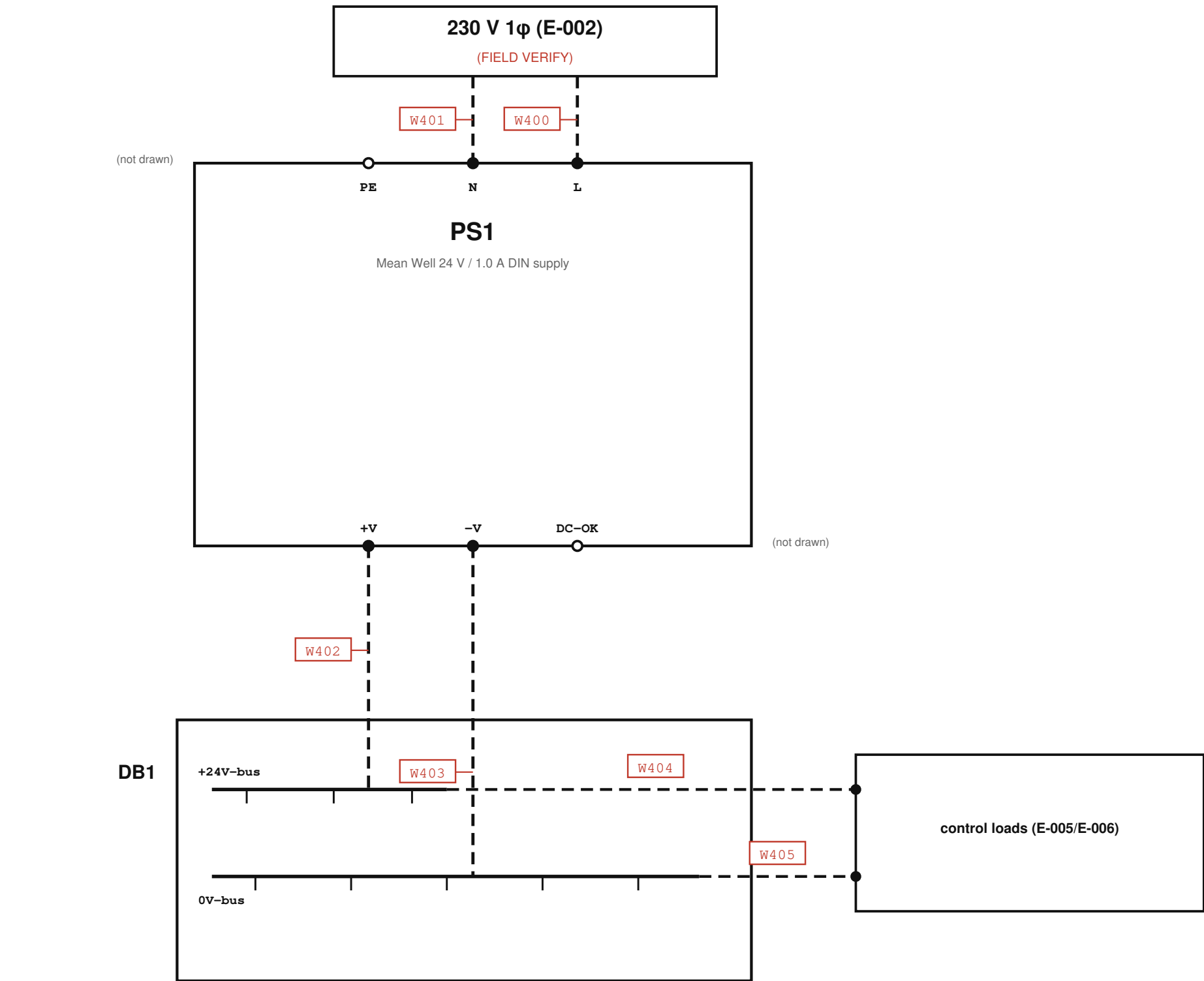
SHEET
E-003

REV A
2026-07-11

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E-004 24 VDC CONTROL POWER

Conv_Simple / CV-101 (garage conveyor) — AC → PS1 (Mean Well 24V/1A) → DB1 +/- distribution block → +24V/0V to E-005, E-006 loads



PS1 is a verified Mean Well DIN-rail supply (24 V / 1.0 A out, 100-240 VAC / 0.55 A in) — nameplate read directly (photo review/photos/wire_2.jpg, 2026-07-11). DB1 polarity (blue vs white = +24V vs 0V), circuit count, PS1 branch fusing, and PS1's AC input source (which leg/circuit feeds L, N) are FIELD VERIFY (OI-25).

SAFETY

- PS1 is fed from AC mains — LOTO + de-energize before opening the panel or metering DB1 (WI p.2 §2).
- Monitored e-stop is NOT a safety stop; de-energizing DB1/PS1 does not remove drive power — see E-003 for the drive supply isolation point.

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- proposed wire number (red = unverified)

CONNECTION TABLE

Wire	From	To	Signal	Type	Status	Notes
W400	230 V 1φ (E-002)	PS1.L	AC line in	power_line	field_verify	
W401	230 V 1φ (E-002)	PS1.N	AC neutral in	power_line	field_verify	
W402	PS1.+V	DB1.+24V-bus	+24 VDC distribution feed	control_24vdc	field_verify	
W403	PS1.-V	DB1.0V-bus	0V distribution return	control_0v	field_verify	
W404	DB1.+24V-bus	control loads (E-005/E-006)	+24V to E-005/E-006 loads	control_24vdc	field_verify	
W405	DB1.0V-bus	control loads (E-005/E-006)	0V to E-005/E-006 loads	control_0v	field_verify	

NOTES

- DB1 is a push-in (WAGO-style), 2-level distribution block fed from PS1's +V/-V output; a bare-copper jumper bridges its top-left terminals in the photo (review/photos/dc_block_full.jpg) — internal jumpering/circuit count not documented.
- PS1's DC-OK terminal (LED/relay status output) and PE input terminal are shown for reference only; neither carries a modeled conductor on this sheet.
- PS1 enclosure/chassis ground and whether DC-0V is bonded to PE anywhere (SELV/PELV scheme vs intentionally floating) is NOT documented — FIELD VERIFY (OI-28).
- +24V / 0V downstream of DB1 feed the E-005 input rail (W24/W0V) and the E-006 output bank 0 feed/return (W600/W609) — same electrical nodes, different sheet.

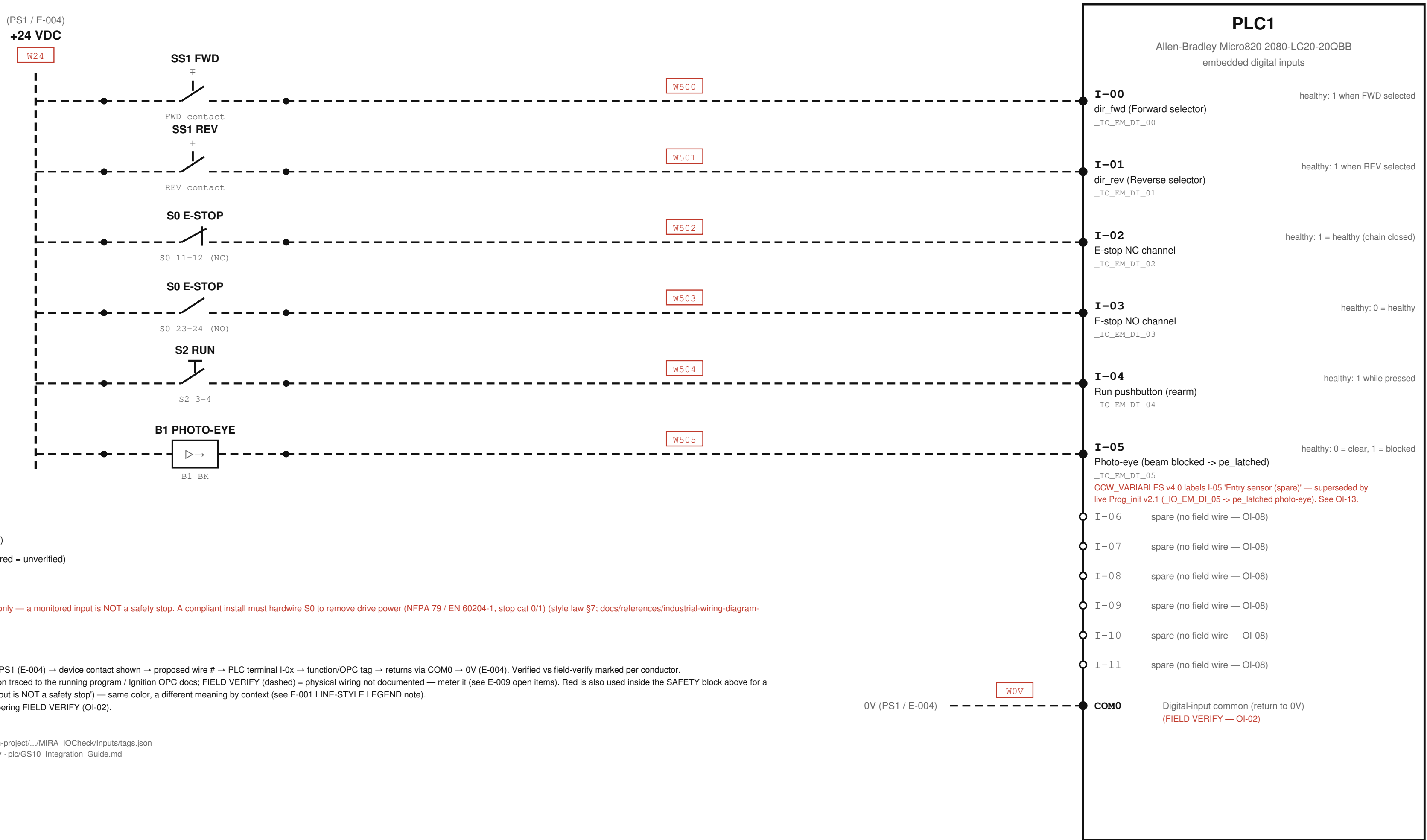
SOURCES:

photo review/photos/wire_2.jpg (2026-07-11 — Mean Well nameplate read directly)
photo review/photos/dc_block_full.jpg + wire_1.jpg (2026-07-11 — DC distribution block)
review/PHOTO_EVIDENCE_V4.md · review/PHOTO_EVIDENCE_V5.md (OI-25)

24 VDC CONTROL POWER Conv_Simple / CV-101 (garage conveyor)	SHEET E-004
Drawn: MIRA / FactoryLM	REV A 2026-07-11 4 of 9

E-005 PLC DIGITAL INPUTS

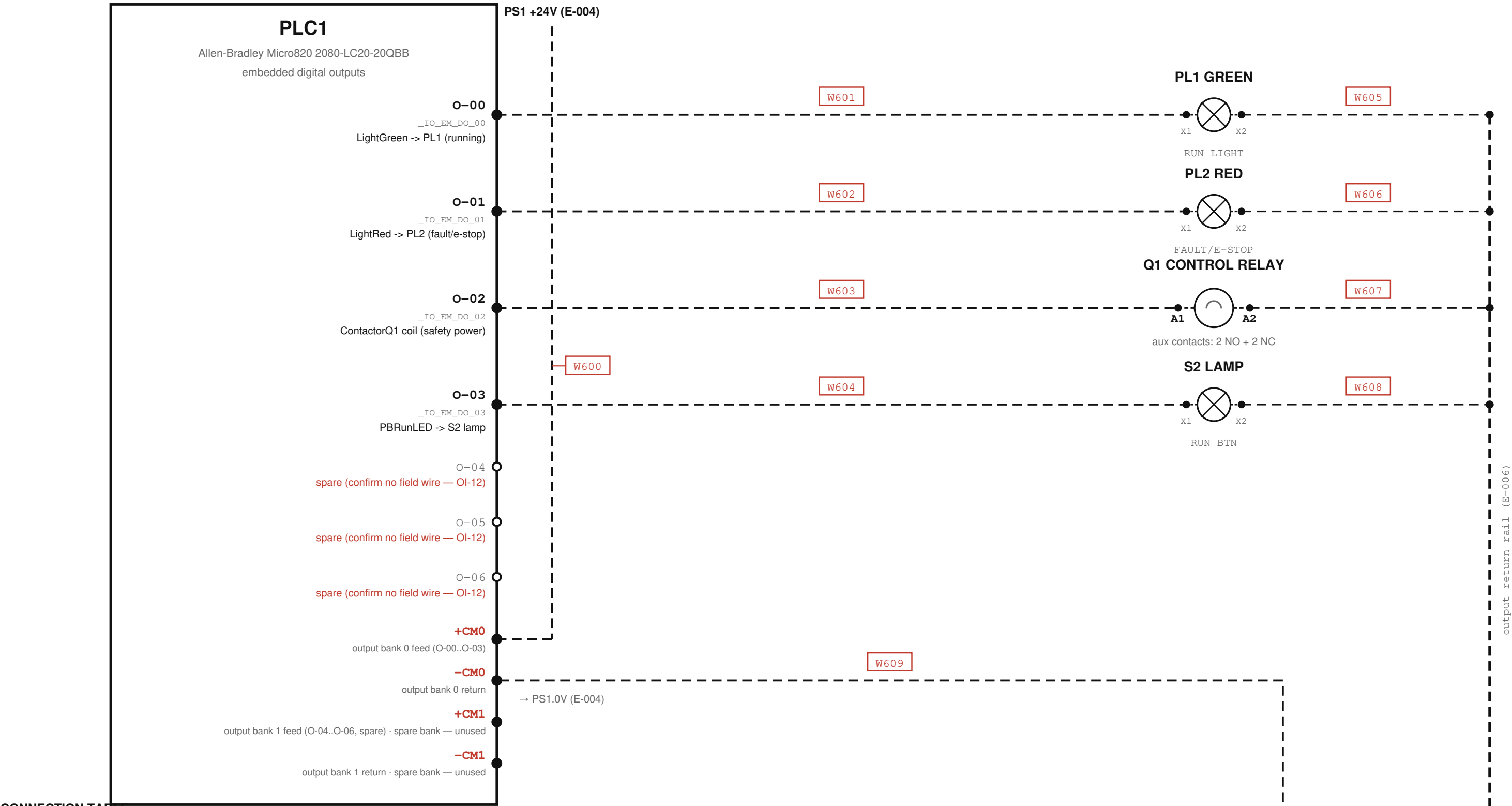
Conv_Simple / CV-101 (garage conveyor) — Allen-Bradley Micro820 2080-LC20-20QBB embedded DI (24 VDC)



PLC DIGITAL INPUTS Conv_Simple / CV-101 (garage conveyor)	SHEET E-005
Drawn: MIRA / FactoryLM	REV A 5 of 9 2026-07-11

E-006 PLC OUTPUTS

Conv_Simple / CV-101 (garage conveyor) — O-00..O-06 · pilots, Q1/MLC control-relay coil, button lamp · GS10 control = HYBRID Modbus + hardwired I/O (E-007; map OI-22)



Hardwired fallback documented in MIRA-WI-001 v3 §3 Fig 1 (DO_03..DO_07 → GS10 DI1..DI5 + DCM) — NOT ACTIVE and NOT drawn: P02.0x at factory default per the 2026-05-20 parameter export; DO_07 does not exist on the 2080-LC20-20QBB (O-00..O-06); DO_03 collides with PBRunLED in the live I/O map. Bench presence of any such wiring = FIELD VERIFY (OI-18).

SAFETY

- Monitored outputs are NOT a safety function.
- E-stop must remove power per NFPA 79.
- Q1/MLC is a CONTROL RELAY (24 VDC coil, driven by O-02) whose 2 NO contacts (13-14, 43-44) switch the drive's 230 V 1φ supply (V6 correction, E-003) — de-energizing O-02 DOES remove Q1's switched supply to the GS10. Q1 is NOT an NFPA-79/EN-60204-1 safety-rated disconnect (small control-relay contacts, not a lockable safety contactor); the e-stop (S0) → O-02 relationship is FIELD VERIFY (OI-26). Do not treat this circuit as the LOTO isolation point — always LOTO at CB1 or upstream.
- O-02 is the sole MLC/Q1 drive-enable coil output — do not reuse it for any other load (MIRA_PLC_WorkInstruction_v3.pdf §3: "DO_02 is already the MLC drive-enable coil and must never be reused").

NOTES

- Q1/MLC = Schneider TeSys CA3KN22BD control relay (2 NO + 2 NC, 24 VDC coil); coil A1(+)A2(-) driven by O-02 (verified from photo wire_3). NO contacts 13-14/43-44 switch the VFD's 230 V 1φ supply (see E-003); 21-22/31-32 NC unused.
- GS10 control (full-res photo): run/dir/speed = Modbus — top FWD/REV/DI connector is EMPTY, RJ45 landed. Bottom analog/output row (+10V/ACM/AI/AO1/DO1/DOC) is wired — likely speed reference + feedback. Map = OI-22.
- NO contacts 13-14 & 43-44 switch the 230 V 1φ VFD supply (see E-003); coil A1/A2 from O-02. NC 21-22/31-32 unused.
- GS10 run/dir/freq = Modbus — see E-007 (P00.20=1, P00.21=2 per 2026-05-20 parameter export)
- No as-built: O-00/O-01/O-03 from CCW v4.0; O-02 corroborated by live Prog_init v2.1.
- Output bank technology + common feed = FIELD VERIFY (OI-09): WI-001 p.4 says relay dry contacts; ±CM polarity naming suggests DC-fed banks — conflicting evidence, resolve per 2080-IN009 + meter.
- GS10 manual warns against cycling a power-circuit switching device for normal run/stop — emergency situations only (GS10_UM.txt L1754-1757, L1811-1813). This applies to Q1/MLC (energized by O-02, switches the drive's 230 V 1φ supply — see E-003): O-02/Q1 stays energized during normal operation; run/stop is commanded via Modbus (E-007), never by cycling Q1.
- S2 lamp (this sheet, O-03/PBRunLED) and S2's run/rearm pushbutton contact (E-005, I-04) are two elements of the SAME physical device — same cross-reference symmetry as Q1/MLC's coil (this sheet) ↔ power contacts (E-003).

SOURCES:

CCW_VARIABLES_v4.0.txt
Prog_init_ConvSimple_v2.1.st
MIRA_PLC_WorkInstruction_v3.pdf
GS10_actual_parameters_5.20.26.xlsx
GS10_UM.txt
photos review/photos/wire_1..4.jpg (2026-07-11, actual devices)
full-res photos review/photos/"_full.jpg + crops (2026-07-11);
review/PHOTO_EVIDENCE_V5.md

LEGEND

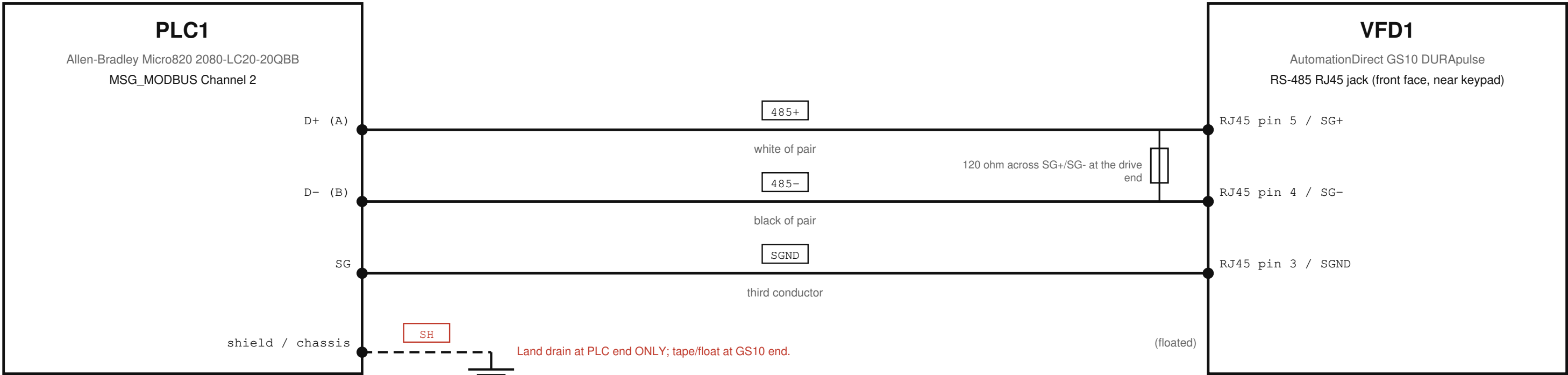
- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxx proposed wire number (red = unverified)

Wire	From	To	Signal	Type	Status	Notes
W600	PS1.+24V	PLC1.+CM0	output bank 0 feed	power_24vdc	field_verify	
W601	PLC1.O-00	PL1.X1	LightGreen (run light)	control_output	field_verify	
W602	PLC1.O-01	PL2.X1	LightRed (fault/e-stop light)	control_output	field_verify	
W603	PLC1.O-02	Q1.A1	ContactorQ1 coil (safety power)	control_output	field_verify	
W604	PLC1.O-03	S2.X1	PBRunLED (run button lamp)	control_output	field_verify	
W605	PL1.X2	output return rail (E-006)	run light return	control_return	field_verify	
W606	PL2.X2	output return rail (E-006)	fault light return	control_return	field_verify	
W607	Q1.A2	output return rail (E-006)	contactor coil return	control_return	field_verify	
W608	S2.X2	output return rail (E-006)	PB lamp return	control_return	field_verify	
W609	PLC1.-CM0	PS1.0V	output bank 0 return	power_0v	field_verify	

PLC OUTPUTS Conv_Simple / CV-101 (garage conveyor)	SHEET E-006
Drawn: MIRA / FactoryLM output map CCW v4.0 + live Prog_init v2.1; O-02 do-not-reuse (MIRA_PLC_WorkInstruction_v3.pdf §3) — see SAFETY	REV A 2026-07-11 6 of 9

E-007 RS-485 / MODBUS RTU COMMUNICATION

Conv_Simple / CV-101 (garage conveyor) — Allen-Bradley Micro820 2080-LC20-20QBB (master) ↔ AutomationDirect GS10 DURApulse (node 1) · Modbus RTU only



CONNECTION TABLE

Src dev	Src terminal	Dst dev	Dst terminal/pin	Cable / conductor	Wire	Evidence	Notes
PLC1	D+ (A)	VFD1	RJ45 pin 5 / SG+	Belden 3105A pair · white of pair	485+	verified	Polarity: if silent/garbled, swap 485+/485- first.
PLC1	D- (B)	VFD1	RJ45 pin 4 / SG-	Belden 3105A pair · black of pair	485-	verified	
PLC1	SG	VFD1	RJ45 pin 3 / SGND	3rd conductor · third conductor	SGND	verified	Dedicated wire, NOT the shield.
PLC1	shield / chassis	VFD1	(floated)	tinned drain · drain	SH	field_verify	Land drain at PLC end ONLY; tape/float at GS10 end.

CCW SERIAL PORT: Modbus RTU, Master · 9600 · 8N1 (P09.04 = 12) · Channel 2 · Node 1 (P09.00 = 1)

Older GS10_Integration_Guide.md said 8N2 (P09.04=13) - SUPERSEDED by bench-verified 8N1.

Line params pending re-verify (OI-20): 2026-05-20 export read 38.4k/8N2; 2026-05-26 bench sniff read 9600/8N1 (shown). Fresh keypad/readback to adjudicate.

COMMAND WORDS (0x2000): STOP = 1 · FWD+RUN = 18 · REV+RUN = 34 · freq register 0x2001

34 NOT 20 — GS10_Integration_Guide.md (2026-03-16) and ControlsToVFD/original PhaseB.st print 20; corrected by PhaseB_V1.4 'FIX 2' (bit 5 REV + bit 1 Run), Beginner_Verify_V2 ('NOT 20'), WI-001 Table 9, and live Prog_init v2.1 (vfd_cmd_word := 34).

Beginner_Verify_V2 §4.4 (GS10 manual p.6); WI-001 Tables 8-10; PhaseB_V1.4.st; Prog_init_ConvSimple_v2.1.st

NOTES

- Belden 3105A — shielded twisted pair (485+/485-) + 1 conductor (SGND) + drain
- VERIFIED (solid) = recovered CommsToVFD + Beginner_Verify p48 + Prog_init v2.1; FIELD VERIFY (dashed) = exact PLC chassis ground point; cable P/N if not Belden 3105A.
- Port locations per CommsToVFD (PLC top-edge connector; GS10 front-face RJ45) — WI-001 says bottom-front. Wayfinding only; confirm on unit (FIELD VERIFY).
- GS10 RS-485 RJ45 cable confirmed landed (photo wire_4). The GS10 control connector (FWD/REV/DI/AI/AO1/DO1) is also wired — see OI-22.
- GS10 RS-485 RJ45 confirmed landed (full-res photo). The GS10 top run/DI control connector is EMPTY — no hardwired run/dir; command is Modbus. Bottom analog/output row is wired (OI-22).
- GS10 run/freq source = Modbus: P00.20=1, P00.21=2 (2026-05-20 export; same fact as E-006).
- P09.09 (Response Delay) = 10.0 ms — program flags CRITICAL (default 2.0 ms → ErrorID-55): Prog_init_ConvSimple_v2.1.st:79.
- 120 ohm termination (SG+/SG- at GS10 end): often omitted on a <2 m bench run — presence is FIELD VERIFY, not confirmed here.

SOURCES:

full-res photos review/photos/*_full.jpg + crops (2026-07-11); review/PHOTO_EVIDENCE_V5.md

DOC REFS:

CCW/MIRA_PLC/docs/instructions/Conv_Simple_CommsToVFD.pdf §2 (recovered style + wiring)
CCW/MIRA_PLC/docs/instructions/Conv_Simple_GS10_Beginner_Verify_V2.pdf p48 (corrected pins/channel)
plc/Prog_init_ConvSimple_v2.1.st (Channel 2, node 1, 0x2103)
plc/GS10_Integration_Guide.md §3 (RJ45 pinout, register map)

TROUBLESHOOTING (this circuit)

- Nothing after config -> swap 485+/485- (A/B polarity; vendors disagree).
- CRC errors -> usually polarity. Complete silence -> usually baud mismatch (confirm 9600 8N1).
- GS10 CE10 / F30 comm-timeout -> no Modbus packets; check wiring, baud, P09.03 timeout.
- 120 ohm termination across SG+/SG- at the GS10 end for long runs (bench <2 m usually fine).
- SGND is a dedicated conductor, not the shield; missing common -> intermittent corruption.

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- proposed wire number (red = unverified)

vfd_comm_ok = TRUE proves the link is up (PLC global).
Read 0x2103 (output freq, Hz x100) via FC03; nonzero readback confirms end-to-end. (Prog_init v2.1:164 'Hz x100'; GS10_UM.txt L15703-05 format XXX.XX Hz)

RS-485 / MODBUS RTU Conv_Simple / CV-101 (garage conveyor)	SHEET E-007
Drawn: MIRA / FactoryLM recovers MIRA-WI-001 / Conv_Simple_CommsToVFD §2	REV A 2026-07-11 7 of 9

E-008 TERMINAL STRIP (X1) + WIRE LIST

Conv_Simple / CV-101 (garage conveyor) — Generated wire list — every conductor across E-003..E-007 (this sheet carries no wires of its own)

WIRE LIST (all sheets, generated)

Wire	From	To	Signal	Sheet	Status
W300	SUPPLY (source — see E-002)	CB1.1	L1 230V 1ph supply	E-003	field_verify
W301	SUPPLY (source — see E-002)	CB1.3	L2 230V 1ph supply	E-003	field_verify
W303	CB1.2	Q1.13	L1 protected -> MLC	E-003	field_verify
W304	CB1.4	Q1.43	L2 protected -> MLC	E-003	field_verify
W305	Q1.14	VFD1.R/L1	L1 switched -> drive	E-003	field_verify
W306	Q1.44	VFD1.S/L2	L2 switched -> drive	E-003	field_verify
W310	VFD1.U/T1	M1.T1	motor phase U	E-003	field_verify
W311	VFD1.V/T2	M1.T2	motor phase V	E-003	field_verify
W312	VFD1.W/T3	M1.T3	motor phase W	E-003	field_verify
W315	VFD1.GND	PE bus	drive PE (≤0.1Ω)	E-003	field_verify
W316	M1.PE	PE bus	motor frame PE	E-003	field_verify
W317	PE bus	SUPPLY (source — see E-002)	PE to source	E-003	field_verify
W400	230 V 1φ (E-002)	PS1.L	AC line in	E-004	field_verify
W401	230 V 1φ (E-002)	PS1.N	AC neutral in	E-004	field_verify
W402	PS1.+V	DB1.+24V-bus	+24 VDC distribution feed	E-004	field_verify
W403	PS1.-V	DB1.0V-bus	0V distribution return	E-004	field_verify
W404	DB1.+24V-bus	control loads (E-005/E-006)	+24V to E-005/E-006 loads	E-004	field_verify
W405	DB1.0V-bus	control loads (E-005/E-006)	0V to E-005/E-006 loads	E-004	field_verify
W0V	PLC1.COM0	PS1.0V	0V / input common	E-005	field_verify
W24	PS1.+24V	+24V rail (this sheet)	+24 VDC	E-005	field_verify
W500	SS1.FWD	PLC1.I-00	dir_fwd	E-005	field_verify
W501	SS1.REV	PLC1.I-01	dir_rev	E-005	field_verify
W502	S0.11-12	PLC1.I-02	e_stop_nc (healthy=1)	E-005	field_verify
W503	S0.23-24	PLC1.I-03	e_stop_no (healthy=0)	E-005	field_verify
W504	S2.3-4	PLC1.I-04	run_pb	E-005	field_verify
W505	B1.BK	PLC1.I-05	photo_eye	E-005	field_verify
W600	PS1.+24V	PLC1.+CM0	output bank 0 feed	E-006	field_verify
W601	PLC1.O-00	PL1.X1	LightGreen (run light)	E-006	field_verify
W602	PLC1.O-01	PL2.X1	LightRed (fault/e-stop light)	E-006	field_verify
W603	PLC1.O-02	Q1.A1	ContactoQ1 coil (safety power)	E-006	field_verify
W604	PLC1.O-03	S2.X1	PBRunLED (run button lamp)	E-006	field_verify
W605	PL1.X2	output return rail (E-006)	run light return	E-006	field_verify
W606	PL2.X2	output return rail (E-006)	fault light return	E-006	field_verify
W607	Q1.A2	output return rail (E-006)	contactor coil return	E-006	field_verify
W608	S2.X2	output return rail (E-006)	PB lamp return	E-006	field_verify
W609	PLC1.-CM0	PS1.0V	output bank 0 return	E-006	field_verify
485+	PLC1.D+ (A)	VFD1.RJ45 pin 5 / SG+	RS-485 inverting (D+ = SG+)	E-007	verified
485-	PLC1.D- (B)	VFD1.RJ45 pin 4 / SG-	RS-485 non-inverting (D- = SG-)	E-007	verified
SGND	PLC1.SG	VFD1.RJ45 pin 3 / SGND	signal ground / common reference	E-007	verified
SH	PLC1.shield / chassis	VFD1.(floated)	cable shield	E-007	field_verify

NOTES

- X1 (field terminal strip, devices.yaml) is where field devices land before the PLC. Its physical terminal-strip layout and numbering are NOT documented in any recovered source — FIELD VERIFY. This table lists the conductors on each side of it, indexed by their owning sheet — it is not X1's internal terminal map.
- Rows are sorted by sheet, then wire number/label. VERIFIED (black) vs FIELD VERIFY (red) mirrors each conductor's status on its owning sheet — see that sheet for the drawn circuit and evidence.

SOURCES:

model/wires.yaml · model/e007_rs485.yaml (generated — do not hand-edit this table; re-run emit after any model change)

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxx

proposed wire number (red = unverified)

TERMINAL STRIP (X1) + WIRE LIST

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

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E-009 OPEN ITEMS / FIELD VERIFICATION

Conv_Simple / CV-101 (garage conveyor) — Generated from open_items.yaml — the formal field-verification docket

OPEN ITEMS (generated)

28 open items; 2 resolved

OI	Sheet	Item	Verify
OI-01	E-005	+24 VDC source terminal on PS1 feeding the input rail	Meter PS1 +24V to 0V = 24 VDC; confirm which terminal feeds the input devices.
OI-02	E-005	Input common (COM0) grouping + sink vs source jumpering on the Micro820	Confirm which embedded input common serves I-00..I-05 and whether inputs are sink or source; meter COM0 to 0V.
OI-03	E-005	All input wire numbers (W200..W205, W24, W0V are PROPOSED)	Read the actual wire markers on the panel; replace proposed numbers with as-built.
OI-04	E-005	Field-device terminal designations (S0 11-12/23-24, SS1 FWD/REV/COM, S2 3-4, B1 BN/BU/BK)	Confirm each device's real terminal markings against its nameplate/datasheet.
OI-05	E-005	E-stop architecture between S0 and the PLC	Confirm whether S0 lands directly on I-02/I-03 or via a safety relay; a monitored input is NOT a safety stop (see safety note).
OI-06	E-005	Photo-eye B1 device (historically 'unmapped') — make/model, PNP/NPN, powered source	Identify the physical sensor; confirm BN/BU/BK to +24V/0V/I-05; PNP sourcing assumed.
OI-07	E-005	Direction selector SS1 physical device (2-pos vs 3-pos FWD-OFF-REV)	Confirm selector type and that FWD/REV contacts map to I-00/I-01.
OI-08	E-005	Spares I-06..I-11 confirmed unused	Confirm no field wires land on I-06..I-11 (program shows none).
OI-09	E-006	Output bank technology + common feeds. +CM0/-CM0/+CM1/-CM1 polarity naming suggests DC transistor banks; devices.yaml io line says 'relay DO' — conflict.	Resolve per 2080-IN009 + meter the bank feed (expect PS1 +24V/0V).
OI-10	E-006	Q1 contactor identity — part number, coil voltage, pole count; coil terminal ids A1/A2 are PROPOSED.	Meter coil circuit from O-02.
OI-11	E-006	PL1/PL2 devices + lamp terminal ids (X1/X2 proposed); S2 lamp terminals.	Meter the light circuits and confirm device markings.
OI-12	E-006	Confirm no field wires on spare O-04..O-06 (+CM1 bank unused).	Trace the PLC output block and verify no wires to O-04, O-05, O-06.
OI-13	E-005, E-006	Re-confirm input AND output map against the live Conv_Simple_1.8 CCW project; v4.0 vintage drift proven on I-05 'Entry sensor' vs live photo-eye.	Open the live Conv_Simple_1.8 CCW project and cross-check every I-xx and O-xx assignment against the drawn functions (I-05 drift already proven; O-02 has independent Prog_init corroboration).
OI-14	E-003	Q1 contactor placement in the power chain (assumed line side of VFD1 per 'safety power'; could be elsewhere).	Trace conductors on the bench; verify Q1 poles 1-6 actual locations. R-C surge absorber both ends recommended per GS10 manual.

OI	Sheet	Item	Verify
OI-15	E-003	Supply voltage/receptacle + CB1 existence/type/rating (2-pole, single-phase 230 V branch breaker).	Identify the bench supply; confirm CB1 device and rating. NEC and GS10 manual require upstream protection.
OI-16	E-003	GS10 exact model+frame (enables wire-gauge/fuse tables); M1 nameplate; conductor gauges; PE points + ground resistance ≤0.1Ω check.	Photograph GS10 nameplate and motor nameplate; measure PE resistance between drive/motor grounds and supply PE.
OI-17	E-003	RFI jumper in/out per grounding topology (symmetrical vs corner-grounded unknown).	Determine bench PE bonding strategy; verify RFI jumpers per GS10_UM.txt L1693-1718 if present.
OI-18	E-006	Fallback DI wiring physical presence (expected ABSENT — P02.0x factory default).	Trace the PLC output block; confirm zero hardwired leads to any GS10 digital inputs (fallback path should be unpopulated).
OI-19	E-003, E-005, E-006	Wire-numbering convention adopted: W[sheet-digit][line-2d] (see wires.yaml convention + the E-001 key). E-005 renumbered W200..W205 → W500..W505. Rails (W24, W0V) and E-007 mnemonic labels (485+/485-/SGND/SH) are exempt. All numbers PROPOSED until as-built.	Read the actual wire markers on the panel; replace proposed W-numbers with as-built (mirror of OI-03).
OI-20	E-007	GS10 comms line params: 2026-05-20 export = 38.4k/8N2 vs 2026-05-26 bench sniff = 9600/8N1 — adjudicate.	Fresh keypad readback of the GS10 serial config; update print if params differ from current 9600/8N1 assumption.
OI-21	E-003	RESOLVED (2026-07-11): no separate motor contactor — the MLC control relay IS the single-phase drive-supply switch (2 NO contacts). See E-003.	Confirm whether a separate motor contactor exists between CB1 and VFD1; the device labeled 'MLC' (Q1) is a Schneider TeSys CA3KN22BD CONTROL RELAY — it has no main power poles.
OI-22	E-006, E-007	GS10 control map (full-res photo): top run/DI connector (FWD/REV/DI3-5/+24V/DCM) confirmed EMPTY; bottom analog/output row (+10V/ACM/AI/AO1/DO1/DOC) confirmed WIRED. Identify which bottom-row terminals are used (speed reference +10V/AI/ACM vs feedback AO1/DO1/DOC) and to what. Confirm nothing on the top run/DI row.	Map which of FWD/REV/DI3/DI4/DI5/AI/AO1/DO1/DOC are landed and to what; reconcile hybrid Modbus + hardwired-terminal control.
OI-23	panel	Siemens CPU 1212C AC/DC/RLY (S7-1200) observed on the panel (photo wire_2) — role vs CV-101 unknown.	Confirm whether the Siemens CPU 1212C controls any CV-101 function before referencing it on any CV-101 sheet; not drawn until evidenced.
OI-24	panel	Device labeled "PMC 192.x" observed on the panel (photo wire_2).	Identify the device (network device / meter?) and its relationship, if any, to CV-101.
OI-25	E-004	DC +/- distribution block: which color (blue vs white) is +24V vs 0V, circuit count, PS1 branch fusing; PLUS PS1's AC input source — which leg/circuit feeds L and N (120 V leg vs the 230 V 1ϕ supply) is not documented.	Trace/meter the distribution block conductors against PS1 +V/-V output; confirm color-to-polarity mapping, circuit count, and any branch fusing (photo wire_1/wire_2). Separately, identify the branch circuit feeding PS1.L/PS1.N and its voltage (single leg of the 230 V 1ϕ supply, or a dedicated 120 V circuit).
OI-26	E-006	MLC (Q1) aux contacts (full-res photo): coil A1/A2 wired + NO contacts 13-14 & 43-44 in use confirmed; NC 21-22/31-32 appear unused. Trace where 13-14 and 43-44 land in the control circuit.	Trace each aux contact to its destination in the control circuit; confirm function (e.g., interlock feedback, seal-in, indicator).
OI-27	E-003	RESOLVED (2026-07-11): motor is 230 V (technician-confirmed) — the GS10 is a standard 230 V 1ϕ-in / 230 V 3ϕ-out drive; the earlier reported 480 V was a misstatement. E-003 letters 230 V 3ϕ output.	Closed by technician confirmation — review/PHOTO_EVIDENCE_V7.md §3 ('it's 230'), not PHOTO_EVIDENCE_V6.md (which predates this confirmation and still shows 480 V/pending). Exact GS10 model/frame still tracked under OI-16 (for wire-gauge/fuse tables).
OI-28	E-004	PS1 enclosure/chassis ground and DC-0V-to-PE bonding not documented — is PS1's DIN-rail/enclosure bonded to PE, and is the 24 VDC 0V rail (DB1 0V-bus) intentionally tied to PE anywhere (SELV/PELV grounding scheme), or left floating?	Meter PS1 enclosure/DIN rail to the PE bus for continuity; meter DB1 0V-bus to the PE bus for continuity/resistance; confirm whether a deliberate 0V-PE bond exists or the 24 VDC system is intentionally floating.

NOTES

- RESOLVED items stay listed (grayed, not deleted) for traceability — see the item text for the resolving evidence and date.
- Closing an open item requires evidence — a repo artifact citation or a bench measurement — not a re-assertion (see the model discipline note on E-001).

SOURCES:

model/open_items.yaml (generated — do not hand-edit this table; re-run emit after any model change)

OPEN ITEMS / FIELD VERIFICATION

Conv_Simple / CV-101 (garage conveyor)

Drawn: MIRA / FactoryLM

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